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Generative AI in engineering education: understanding acceptance and use of new GPT teaching tools within a UTAUT framework

Christopher Honig [a], Shannon Rios [b] and Aditya Desu [c]

[a]Department of Chemical Engineering, Faculty of Engineering and IT, Melbourne, Australia; [b]Teaching and Learning Lab, Faculty of Engineering and IT, Melbourne, Australia; [c]Faculty of Engineering and IT, The University of Melbourne, Melbourne, Australia

ABSTRACT

The proliferation of generative AI (genAI) has sparked debate on its integration into university curricula, particularly in engineering education. This paper examines the adoption of genAI-based teaching tools in an undergraduate chemical engineering course, within the theoretical scaffold of the Unified Theory of Acceptance and Use of Technology (UTAUT). We investigate the voluntary use of custom GPT-powered chatbots, designed to mirror simulated industry consultants and viva/defence interviews within engineering safety case study activities. Through an exploratory mixed-method approach, utilising student surveys and tracking direct software usage, we have identified high student uptake of the AI-tools, implying effective integration into the student learning practices. We identify Performance Expectancy as the most significant factor influencing usage, with concerns around the AI-tools' accuracy and scope functioning as potential barriers to wider uptake. Interestingly, a subset of students expressed a preference for AI teaching tools over traditional in-person interactions, indicating an overlooked potential for AI in addressing social anxiety as a barrier to learning, for some students. This offers an exciting new avenue for adaptive and personalised learning experiences, to enhance student engagement, access and success.

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Chemical engineering; engineering education; safety; case study; industry; consultant; generative AI; GenAI ChatGPT; UTAUT; technology acceptance

1. Introduction

Following the release of ChatGPT in late 2022, there has been an explosion of interest in genAI for use in education (Lo 2023). The potential benefits are widely recognised for personalised learning, round-the-clock access for students and enhancing accessibility, particularly for students with disabilities or language barriers (Mosaiyebzadeh et al. 2023). There are also new opportunities for interactive and engaging learning environments (Adeshola and Adepoju 2024; C. Honig 2024a; C. D. Honig, Rios, and Oliveira 2023; Jacobsen and Weber 2023).

To date, much of the engineering education literature has focused on the 'risks and opportunities' of AI within Universities (Mosaiyebzadeh et al. 2023; Neumann, Rauschenberger, and Schön 2023). This is reasonable when considering the relative infancy of this technology, but it does mean the conversation to date has been largely theoretical, discussing *potential* pitfalls or *possible* benefits. There is a complementary need to trial, test and evaluate teaching innovations that leverage this technology, and to map these to established theoretical models of learning technology adoption to better understand the barriers to technology acceptance in classrooms. At the same time, it is also important to create and disseminate these new teaching strategies and resources to other educators to stimulate teaching and learning innovation and experimentation with this disruptive technology.

In this study, we seek to understand how genAI teaching innovations are perceived and used by students in the context of a chemical engineering subject at the University of Melbourne. While student perceptions of generic ChatGPT have been studied previously (Ngo 2023; Shoufan 2023), there have been limited studies on the use of customised GPT-powered teaching tools, and barriers to their broader uptake. To support this investigation, we have implemented two similar genAI learning activities: 1. To roleplay as an industry consultant for students to interview, in the context of a well-known process safety incident and 2. To roleplay as an academic conducting a viva to assess the student's understanding of the process safety incident. Both activities are designed to support student learning and to help them prepare for interacting with real humans (i.e. industry consultants and academic assessors).

The unified theory of acceptance and use of technology (UTAUT) framework is used as a lens to understand student reflections and feedback related to their participation in genAI learning activities. This study builds on previous work that explored the design, implementation and use of genAI learning activities. This work showed how AI used to roleplay as industry consultants, can offer

CONTACT Christopher Honig christopher.honig@unimelb.edu.au Department of Chemical Engineering, Faculty of Engineering and IT, University of Melbourne, Melbourne, Australia

a new low-risk teaching format for students who struggle with speaking up and may fear 'losing face' in front of instructors or peers (C. D. Honig and Desu 2024; C. Honig, Desu, and Franklin 2024).

2. Background

2.1. GenAI

GenAI is a sub-branch of artificial intelligence (AI) that refers to AI models that can produce unique and original works related to the data they were trained on (García-Peñalvo and Vázquez-Ingelmo 2023). In particular, genAI models are able to create human-like, high-quality material including images, text, sound and video. Large language models (LLMs), such as ChatGPT, are genAI models designed specifically to generate textual data with text-based input.

Regardless of how academics feel about the future use of AI in tertiary education, the reality is, it is now here and will become increasingly integrated into the structure of education. Students are able to use it in a wide range of ways, both legitimate and malicious, that may not be detectable (Gallent Torres, Zapata-González, and Ortego-Hernando 2023; Uzun 2023). Academics can also leverage genAI in their work to design learning activities or use it themselves to improve the efficiency and quality of their work.

To date, much of the discussion within the literature around genAI in teaching has focused on concerns around academic integrity (Eke 2023; Sullivan, Kelly, and McLaughlan 2023) although contract cheating has presented very similar threats to academic integrity in the past, the rise of genAI has ultimately only increased accessibility and equity of these forms of cheating. While noting this historical caveat, genAI still presents a significant risk to assessment in higher education due to its accessibility and difficulty to detect. While there are currently some shortcomings in terms of performance, genAI and, in particular, large language models have shown their ability to rapidly improve over the last year. Additionally, as user skill improves, the list of 'at-risk' assessment types has also increased (Nikolic et al. 2024).

While educators will always show interest in exploring new teaching strategies, disruptive technologies like genAI inevitably evoke mixed reactions among users – in this case, students and teachers. Some may view the technology as merely a passing trend with limited effectiveness, while others might regard it as a transformative solution to many of their challenges (Onal 2023; Rasul et al. 2023). If we want to leverage this technology, then it will be important to understand how the technology is accepted and used by academic staff and students.

The teaching use-cases for genAI have grown dramatically over the past 2 years, and it can be difficult to keep on top of all the new work, given the explosion of academic interest (Bahroun et al. 2023). Current uses of genAI focus on personalised and interactive learning, and the generation of formative assessment activities to provide continuous, iterative feedback (Baidoo-Anu and Ansah 2023; C. Honig 2024a, C. Honig 2024b). In a use-case similar to our own, genAI is being used to roleplay as a patient for diagnosis for medical students (Preiksaitis and Rose 2023; Sardesai et al. 2024). This is similar to our first use-case as the GPT roleplays as a person, so the design and insights are often directly transferrable; we adopted this learning design directly and it evidences the value of cross-disciplinary teaching practice; GPTs enable new teaching practices that may not have been possible previously (due to cost or scale). A key advantage of AI-roleplay applications is their lower standards of informational precision, as the GPT roleplays as an unreliable person rather than infallible tutor (C. D. Honig and Desu 2024).

In our second use-case (a simulated Viva for formative feedback and practice for summative assessment) we have not yet found comparable work within the literature, although we note that GPTs are being widely used to provide formative feedback on student writing (Zhang et al. 2024) and vivas are commonly suggested as an assessment method to defeat GPT-ghostwriting (Moorhouse, Yeo, and Wan 2023; Pearce and Chiavaroli 2023).

2.2. Unified Theory of Acceptance and Use of Technology (UTAUT)

A useful theoretical scaffold for understanding technological acceptance within organisations is the UTAUT (Chang 2012; Dwivedi et al. 2019; Venkatesh et al. 2003; M. D. Williams, Rana, and Dwivedi 2015). It is a comprehensive and widely applied theoretical model, used in technology acceptance research. It integrates elements from various existing theories and models including the Technology Acceptance Model (TAM), the Theory of Planned Behaviour (TPB), and the Social Cognitive Theory (SCT). Although it has been widely cited in the literature, the theory and its constructs are rarely used in direct empirical research (M. Williams et al. 2011) implying it can offer new insights when applied proactively.

The UTAUT frames technology acceptance both through the conditions of the technology and context (such as the software itself and the University Facilitating Conditions) but also through respondents' *perceptions* of the technology. Technology acceptance and use is primarily understood through four constructs:

(1) Performance Expectancy: The degree to which the individual believes the new technology will improve their job performance, for example, how the student perceives that an AI tool can improve their learning outcomes.
(2) Effort Expectancy: The perceived ease of learning the new technology, for example, how easy a student thinks an AI tool will be to use.
(3) Social Influence: How strongly the individual perceives important others (such as peers, subject coordinators, graduate employers) believe they should use the new system.
(4) Facilitating Conditions: Presence of organisational and technical infrastructure

To support the use of the system and can include available resources, appropriate training, knowledge and support.

Constructs 1–3 are all based on user *perceptions* of the technology rather than specific features of the software or institution itself. Construct 4 focuses on the institutional context the technology is to be used in. The UTAUT framework posits that these four constructs are direct determinants of behavioural intention (the intention to use the technology) and the use behaviour (actual usage of the technology). These four constructs are in turn moderated by gender, age, experience, and voluntariness of use. By examining these constructs in a classroom setting, we can identify key influences to acceptance of a technology. This provides not only a framework for understanding the perception of the tool (driving use-behaviour) but also the performance of the tool (through long-term acceptance).

We note that since its original inception, there have been a number of extensions to the UTAUT, most notably the UTAUT2 (Venkatesh, Thong, and Xu 2012) which includes the further constructs of 'Hedonic motivation', 'price value' and 'experience and habit', which emphasises intrinsic motivations for using and accepting technology. Our study has a greater interest in extrinsic motivations, as these are within the remit of the teaching facilitator. Although intrinsic motivations are interesting, an academic must teach to all students within their class, regardless of personal motivation. Ultimately, all facilitator-led strategies to improve student engagement are necessarily extrinsic motivators to the student.

Within the literature, just as ChatGPT has seen rapid growth in its user base, there has been corresponding growth in studies about genAI. We note a number of other studies that have used the UTAUT theoretical framework, to understand the use of ChatGPT specifically. The UTAUT has been used with the further constructs of 'Perceived Interactivity' and 'Privacy Concerns' to understand user adoption through semi-structured qualitative interviews (Menon and Shilpa 2023). Similarly, it has been extended with 'Relative risk perception' to understand acceptance and use of ChatGPT with data collected from an online panel (Lee et al. 2024). Both studies apply the UTAUT to 'general' users of ChatGPT, but the model has also been applied to study university students specifically, in work more directly related to our own (Budhathoki et al. 2024). This work uses the additional construct of 'anxiety' to understand acceptance of the technology.

2.3. Context to our study: safety case studies

The subject where the genAI tools are implemented uses chemical engineering safety case studies as a tool to facilitate learning. Process safety has become a paramount focus in engineering education, with a shift towards industry-facing programmes and the desire to align curriculum to the contemporary workplace needs of student graduates (Amaya-Gómez et al. 2019; Meyer 2017; Mkpat, Reniers, and Cozzani 2018; Perrin et al. 2018). Safety concepts are most commonly taught through the study of historical process safety incident case studies, because these can offer direct real-world applications, technical understanding of underlying safety issues and naturally introduce ethical considerations and professional responsibilities (Guntzburger, Pauchant, and Tanguy 2017; Martin, Conlon, and Bowe 2021; Qian, Vaddiraju, and Khan 2023; Sandri et al. 2023). However, recent work has emphasised the link between effective engineering safety education and the safety experience and qualifications of the teaching staff themselves (Kerin and Pollock 2019; Pollock and Sorensen 2021). This suggests that safety can best be taught through integrated engineering programmes (Pollock and Sorensen 2021) and can be significantly enhanced with support from industry consultants with direct experience with safety in industry (C. Honig, Desu, and Franklin 2024). For example, recently retired engineers can offer significant intellectual capital to engineering programmes: a career of experience, insight into industry culture and intimate technical understanding. The key challenge to this approach is how to balance limited budgets and finite staff availability, against the growing student desire for access to the knowledge bank of teaching staff and industry consultants.

3. Methodology

3.1. Activity design

In 2023, we created custom GPT-powered chatbots to supplement student understanding of case studies for a 3rd year undergraduate subject in chemical engineering called 'Safety and Sustainability Case Studies'. The chatbots used the GPT-3.5 turbo LLM that underpins

ChatGPT (later upgraded to GPT-4). The chatbots were used in two activities:

(1) In the first activity, the chatbots were designed to mimic a real industry consultant, to allow for independent and asynchronous interaction. Students were able to use the chatbot before the consultation, to better understand the consultant's background or address simple questions. After the consultations, students were able to extend discussions with the chatbot to glean additional insights about the safety case studies under investigation and ask additional follow-up questions that could not be covered in constrained consultation time. To build the chatbots, the consultants were first asked to complete a detailed questionnaire, which was then refined, for a single system prompt. Students were instructed to use this chatbot as an experimental research tool and an alternative source of information about the safety case study.
(2) In the second activity, a chatbot was designed to mimic an academic assessment in a viva/oral defence assessment. In the assessment, student teams present their safety case study to a panel (an academic and industry consultant). The chatbot is designed to provide sample questions and feedback on answers to the students. Students were encouraged to use this chatbot in their team, as a focal point for practicing for their viva and team-working dynamics. The chatbot is designed to familiarise students with the assessment format (most of them have not participated in a viva before), provide a focal point for discussion and also provide a mechanism for review of the learnt material from their safety case study. It is primarily intended as a learning activity.

Students were instructed how to use these tools in class and received additional lessons on genAI literacy. A prior study outlines the design, performance and efficacy of the chatbots in more detail (C. Honig, Desu, and Franklin 2024).

An essential point that distinguishes these AI-consultants from other AI-based tutoring, is that the AI-consultants are not designed to provide a font of 'infallible truth' as might be expected from a textbook or a teacher, where there would be no expectation of significant technical errors. Although many students might be aware that genAI can produce inaccuracies (C. Honig, Desu, and Franklin 2024) there is a student expectation that University provided teaching tools are reliable, and consequently that an AI-tutor should perform at a level equivalent to a human tutor. In our context, the chatbots are meant to simulate *consultants*, senior and experienced colleagues in the workplace, but who should not be assumed to be infallible. This creates two important distinctions from other AI-based teaching tools, which can help resolve issues with 'hallucinations' (responses that are factually incorrect or nonsensical):

(1) Not compellable. The chatbot is not required to approximate a correct answer to every query. Within the system prompt, it has been instructed to answer 'I don't know' for questions that are outside of its knowledge bounds. It has also been prompted to answer 'That is outside the scope of the meeting' for questions that are beyond a specific line of query expected for the interaction. This helps suppress hallucinations and helps the chatbot behaves more like a human, who may be uncertain, even within their field of expertise.
(2) Not infallible. Students are specifically prepped to treat the chatbot as equivalent in capability to their industry consultant and not like an omniscient tutor. So we emphasise a distinction between 'highly capable' and 'infallible'. We found this approach to be useful in developing the AI literacy of students, enabling them to critically evaluate and judge the outputs of AI, but it also has broader benefits in developing students' general information literacy skills. Within the workplace, everyone has shared responsibility for safety, so a student engineer should not do something they believe to be unsafe, simply because they have been told to do so by a more senior colleague. Junior engineers should be critically evaluating the information presented to them and not blindly accepting the opinions of experts or more senior colleagues. We seek to replicate this dynamic here as well.

3.2. Research questions

In this study, we seek to answer the following research questions:

(1) Do students perceive a genAI-based teaching activity to be useful?
(2) What specific factors influence voluntary student uptake of a genAI-based teaching activity?

To address these research questions, the study applied the UTAUT framework to understand technology acceptance within a cohort of students (Wang, Hung, and Chou 2006). Given the open-ended nature of the research question, the methodology employed was an exploratory mixed-method approach. Students were engaged in informal discussion about the learning activity and their thoughts on the use of AI more broadly in engineering education. We first undertook a beta testing phase prior to using the chatbots in the subject whereby we observed students interacting with the chatbots. Based on these initial observations and discussions with students, a survey instrument was developed to gauge student reactions across the four constructs of the UTAUT. The behavioural domains within the UTAUT, such as intentional and final use of the software, were also investigated by a survey and validated by cross-checking against recorded use of the software; we did not log individual users but could log the number of unique IP addresses accessing the tool, and then cross-check this to the survey response rates indicating use of the software.

3.3. Survey design

We used a survey method to collect data to help inform answers to our research questions. Surveys have been widely used in UTAUT research (M. D. Williams, Rana, and Dwivedi 2015) and so the findings are more generalisable and transferable to other studies. It is popular within UTAUT research (and informs our research method design) because it offers wide reach and accessibility to the cohort (students do not need to attend class to participate in the study and can work remotely), it can collect both qualitative and quantitative insights and it allows for anonymity, creating more room for honesty. The survey method can also be validated more reliably, as prompt statements from previously validated UTAUT surveys can be reused directly (Herrero, San Martín, and Garcia-De Los Salmones 2017; Wang, Hung, and Chou 2006; M. D. Williams, Rana, and Dwivedi 2015).

For the quantitative component of the survey instrument, a standardised 5-point Likert scale was used to collect responses to prompt statements. The survey consisted of 25 questions related to the teaching activities in the subject, however, only 12 questions related to this research question. As a generalisation to other work, the five options were standardised to the most widely used Likert structure: strongly agree, agree, neither agree nor disagree, disagree, strongly disagree. An option for (not appropriate/did not use) was also made available where needed. Twelve questions used are shown later in the results section alongside a rich discussion.

The qualitative part of the survey featured two questions which were designed to solicit further semi-structured written feedback on the two genAI activities. The feedback was later thematically analysed (discussed further below) with respect to the four UTAUT constructs and the behavioural domains. The questions used were:

(1) ‘Did you make use of the AI consultant chatbot? If not please explain why. If you did, please outline how useful you found it’, and
(2) ‘Did you make use of the AI-Lecturer Chatbot? If not please explain why, if different from above. If you did, please outline how useful you found it’.

Additionally, during the initial interviews and when observing students using the AI-tools during beta-testing, we discovered a subset of students who seemed to have a preference for the AI-tools over real in-person meetings with the consultants. This was a surprising result because we originally conceived the AI-tools as a helpful but inferior supplement to in-person meetings. Due to these observations, the survey was augmented to measure this aspect.

3.4. Data analysis

The quantitative data collected was analysed by first converting the descriptive responses to numerical values (strongly disagree at 1 to strongly agree at 5) and then for each prompt statement calculating the average and percentage of agree/disagree. We note that it is a common practice to present average and standard deviations when interpreting Likert results, but we preferred to look at the percentage agree/disagree for each question, as in some cases, the distribution of responses to the prompts was not a normal distribution. Using a percentage of agree/disagree also helps clarify the subtlety that Likert scores are not linear (for example, a 4 = Agree is not twice as positive as a 2 = Disagree).

For our qualitative data, collected in two open-ended questions, there are a range of possible approaches for analysing these data, including grounded theory, framework method, narrative analysis and phenomenology. For this study, we decided to conduct a thematic analysis (Castleberry and Nolen 2018; Thomas and Harden 2008; Vaismoradi et al. 2016) due to its ease of use to be adapted across a wide range of theoretical frameworks, the relative familiarity the typical reader has with the analysis method, the small number of responses ($n = 41$) and for consistency and generalisability to other related studies. A well-understood limitation of thematic analysis is that it can be less theoretically driven than other approaches like Grounded Theory or Discourse Analysis and is thus more prone to subjectivity and bias (Nowell et al. 2017); thematic analysis tends to give voice to the dataset in an accessible manner rather than shape a response to specific structured research questions. So while this can constitute an advantage for the exploratory focus of the project, we also acknowledge that this can limit the theoretical rigidity of the analysis method.

Table 1. A list of common themes, associated UTAUT construct and frequency of occurrence.

Themes	UTAUT Construct Mapping	Sentiment	Frequency
AI lacking	Performance Expectancy	Neg	19
AI wrong	Performance Expectancy	Neg	2
AI mistrust	Performance Expectancy	Neg	19
Untransferable	Performance Expectancy	Neg	3
Supports preparation	Performance Expectancy	Pos	25
Provides Clarity	Performance Expectancy	Pos	6
Builds understanding	Performance Expectancy	Pos	15
Instructions not clear	Facilitating Conditions	Neg	8
Information verification	Effort Expectancy	Neg	4
Fast response	Effort Expectancy	Pos	2
Unintuitive	Effort Expectancy	Neg	1
Wasn't useful	Behavioural Perception	Neg	11
Was Useful	Behavioural Perception	Pos	21
Prefer other	Other (Preference)	Neg	13

A standard thematic analysis approach was undertaken. The written comments were first uniformly formatted, collated and anonymised. At the same time, we undertook a familiarisation phase by reading through the comments in totality several times. Next, we developed some initial codes to systemically apply to the dataset. We developed these codes by identifying recurrent concerns in the written comments, typically pulling specific phrases from the student response, to summarise the response sentiment. We then iterated over the data several times to map to the codes, while also adding additional codes as we went.

Having developed a large library of codes ($n = 33$) from the data set, we began to group these codes into themes before giving indicative names to the themes. As a final step, once all the themes ($n = 14$) were identified, we connected the themes to explicit constructs within the UTAUT or assigned them an 'Other' theme. Most themes were directly mapped to the Performance Expectancy construct. The themes identified, their corresponding UTAUT construct and frequency of occurrence are shown in Table 1. We also categorised each code and theme as either representing a positive or negative sentiment. We note that the frequency of some themes is low (as few as 1 or 2 responses) but have included them in the analysis since they capture unique and important aspects of students' perceptions towards the technology.

3.5. Survey validation

When collecting data for evaluation, it is important to ensure that any survey instruments are validated. This helps ensure reliability over time (the results are reproducible), internal consistency and accuracy (survey items measure what they are intended to) and provides test–retest reliability (respondents offer similar results at different times). It also ensures validity of the instrument including content validity (it comprehensively covers the topics or construct it is intended to measure), construct validity (it accurately measures the broader construct itself, here the constructs of the UTAUT) and criterion validity (for example, there is correlation and internal consistency between questions of the same construct). Validation also lends credibility to the instrument and allows for generalisation and comparison with other results.

To validate our survey, we began by reviewing similar studies that made use of validated survey instruments that could be immediately applied to the current study. We required that these surveys were already in the public domain and available for use and that the Likert prompt statements and qualitative questions were appropriate for our context. Papers that were included for review were those looking at similar research questions and that also provided strong evidence of the reliability and validity of their survey instruments (Wang, Hung, and Chou 2006). From this review, we found two papers (Cady, Fortenberry et al. 2009, Chan, Zhao et al. 2017) that had suitable question structures to use in our survey.

Prior validated UTAUT surveys were a useful starting point, but they did not cover all question themes appropriate for data collection for our central research questions and the relative novelty of the teaching tool. Additionally, these questions did not allow for test–retest consistency, which was a concern for the researchers. To address this, we developed a number of new bespoke questions, particularly for the open-ended qualitative prompt statements to draw out student comments related to the learning tool. Examples included an assessment of overall perception of learning utility (see section 5.1) and questions related to human vs AI preference (see section 4.5). We took the complete list of prompt statements for the questionnaire and reviewed them all for relevance and comprehensiveness of the study's constructs. The survey was reviewed by colleagues with specific expertise in engineering education and survey validity, as well as the authors of the paper and other educational researchers. The survey questions were checked for cultural and linguistic appropriateness, for general readability and clarity. Finally, the survey was pilot tested with a select group of students (not in the final study). We showed the survey to the students to collect feedback on how they interpreted the instructions and what they were trying to represent when presenting their answers. This process led to several adaptations of the survey instrument but did not significantly change the focus of the survey. The reliability of the survey instrument is confirmed and demonstrated with examples at the end of the results section.

Table 2. Likert questions related to performance expectancy.

Prompt statement	Mean Likert Score (1–5)	Agree/Disagree percentages
4.1.1 Using ChatGPT helped me complete complex tasks in chemical engineering more easily	4.0	Agree = 72.5% Disagree = 7.5%
4.1.2 I have a good understanding of the limitations of ChatGPT, especially in technical subjects.	4.58	Agree = 97.5% Disagree = 0%

3.6. Survey distribution

The survey instrument was distributed to students through an LMS announcement, with a request for voluntary completion. In both the LMS advertisement and plain language statement at the start of the survey, it was made clear that completion of the survey was voluntary. Students were offered the chance to win one of the two gift cards for completing the survey.

Of the 70 students within the class, the survey received 41 complete responses (52 responses in total, but a number was incomplete). This represented an effective response rate of 59% and gives some confidence that the sample set is representative of the cohort experience.

4. Results

Overall, there was a very high reported use of the AI-consultants and AI-Viva/Defence tools. Among the 41 respondents:

- 34 reported using the AI-consultant and 7 reported not using it (83% voluntary uptake among respondents)
- 29 reported using the AI-Viva/Defence and 12 reported not using it (71% voluntary uptake among respondents)

The highly reported use can also be cross-checked against the actual use of the software: we recorded IP addresses accessing the platform and found 78 unique users (of a class of 70). Four correspond to teaching staff. Students may access the platform multiple times across different devices (for example, a laptop and tablet device) accounting for the final total greater than the total class enrolments. This is indicative of very high uptake and re-use across the student cohort, which supports the self-reported uptake findings. We note that the sample size is small which will increase the standard error or lead to p-value inflation and may create sample biases or loss of precision.

4.1. Performance expectancy

The thematic analysis showed that Performance Expectancy was by far the most prominent UTAUT construct represented in students' written responses ($n = 89$). A large volume of the comments focused on performance as a barrier to adoption, shown in Table 1 as a negative sentiment ($n = 43$). The specific themes that were mapped as potential barriers to technology uptake include: 'AI Lacking', 'AI Wrong', 'AI mistrust' and 'Untransferable'. Indicative comments from students in these themes include: '*Felt that use of the chatbot was risky as it might make me learn false details of the case study which would have a bad impact on the assessment*'.

Just as Performance Expectancy can function as a barrier to technological uptake, it can also serve as a driver to engagement. This is represented by the positive sentiment Performance Expectancy aligned themes ($n = 46$) 'Supports preparation', 'Provides Clarity', and 'Builds Understanding'. These include comments such as: '*I found it useful in terms of preparation for the final meeting and to verify any minor information that might have passed my head regarding the case*'.

The Likert question responses, shown in Table 2, indicate a highly positive sentiment towards the capabilities of ChatGPT to improve their performance with only 7.5% of the respondents disagreeing that ChatGPT helped them complete tasks. Students also indicated high confidence in their understanding of ChatGPT capabilities with 97.5% agreeing with the statement '*I have a good understanding of the limitations of ChatGPT, especially in technical subjects*'. This is somewhat counter to the results of the thematic analysis which may show an over confidence in students' proficiency with the tool or a disconnect between how students use ChatGPT generally, versus how they view the AI tools trialled in the subject.

This indicates that the performance of the tool and student perceptions of performance emerge here as key drivers for both student uptake and effective deployment of tools within a teaching context. One option is to try to design more factually reliable GPT-tools, but another strategy involves shifting the educational focus from a 'reliable tutor' to a roleplay, such as patient roleplay for medical diagnosis (Sardesai et al. 2024) or as an engineering consultant in a real-world context (C. Honig, Desu, and Franklin 2024).

4.2. Effort expectancy

Effort Expectancy aligned to three themes ($n = 7$); 'Information Verification' and 'Unintuitive' highlighted negative sentiment, while 'Fast Response' ($n = 2$)

Table 3. Likert responses related to effort expectancy.

Prompt statement	Mean Likert Score (1–5)	Agree/Disagree percentages
4.2.1 Learning how to use ChatGPT for chemical engineering tasks was straightforward.	4.03	Agree = 87.5% Disagree = 7.5%
4.2.2 I can easily integrate ChatGPT into my study routine for chemical engineering.	3.98	Agree = 67.5% Disagree = 5.0%

represented positive sentiment. The 'Information Verification' theme contained the largest number of responses ($n = 4$) and represents responses where students felt they needed to do extra work to verify the AI output and this theme often overlapped with negative Performance Expectancy.

The Likert responses related to Effort Expectancy, shown in Table 3, indicated that students found the instructions easy to follow (Q4.2.1 87% agree), but fewer students found it easy to integrate ChatGPT into their normal study routine (Q4.2.2 67.5% agree). This aligns with the thematic analysis' findings that Effort Expectancy was not a significant barrier but had some impact related to workflow.

4.3. Social influence

The thematic analysis did not identify any themes that aligned with the Social Influence construct which may indicate that there were no significant barriers or enablers to technology acceptance related to the Social Construct or could indicate that Social Construct did not rate highly to students as being important.

Table 4 shows the Likert responses relating to Social Influence. These results indicate that while students generally felt supported by their peers and the institution to use ChatGPT, they do indicate a perception that their peers are more supportive than institutions on the use of genAI.

4.4. Facilitating conditions

The 'Instructions not clear' theme ($n = 8$) was the only one linked to the Facilitating Conditions construct, referring to learning activity instructions rather than AI tool instructions. Comments indicated that students felt unprepared or 'forgot [the tool] existed till the night before', particularly for the AI-defence activity ($n = 6/8$). This could be addressed by providing earlier access and clearer communication about the tool's purpose and benefits. Overall, Facilitating Conditions did not present a significant barrier to technology acceptance, consistent with the tool's simple LMS link and familiar chatbot interface requiring no new skills.

Table 5 shows that students generally did not see access as a barrier, though 20% reported insufficient instructor support. This can be mitigated by extending lead time for preparation and practice. The thematic analysis supports these findings.

4.5. Other observations

Two other themes were identified that do not directly related to any of the four UTAUT constructs. The first of these themes was named 'Behavioural Perception' ($n = 32$). This theme relates to students' perception of the utility of the learning activity, indicating either that they felt it was ($n = 21$) or wasn't ($n = 11$) useful. We felt that this theme was a key indicator of behavioural intent to use the technology in their learning.

Table 4. Likert responses related to social influence.

Prompt statement	Mean Likert Score (1–5)	Agree/Disagree percentages
4.3.1 My peers think that using ChatGPT is beneficial to chemical engineering studies.	4.13	Agree = 85% Disagree = 2.5%
4.3.2 The faculty in my department encourage the use of ChatGPT for learning chemical engineering.	3.90	Agree = 70.0% Disagree = 5.0%
4.3.3 I feel that using ChatGPT is well-regarded among my classmates.	4.13	Agree = 87.5% Disagree = 2.5%
4.3.4 My learning environment supports the use of ChatGPT for educational purposes.	3.90	Agree = 72.5% Disagree = 10%

Table 5. Likert responses related to facilitating conditions.

Prompt statement	Mean Likert Score (1–5)	Agree/Disagree percentages
4.4.1 I have access to the necessary resources (like internet, devices and subscriptions) to use ChatGPT effectively.	4.13	Agree = 80% Disagree = 5%
4.4.2 There is sufficient support (from instructors or technical staff) for teaching ChatGPT in my chemical engineering course.	3.63	Agree = 60% Disagree = 20%

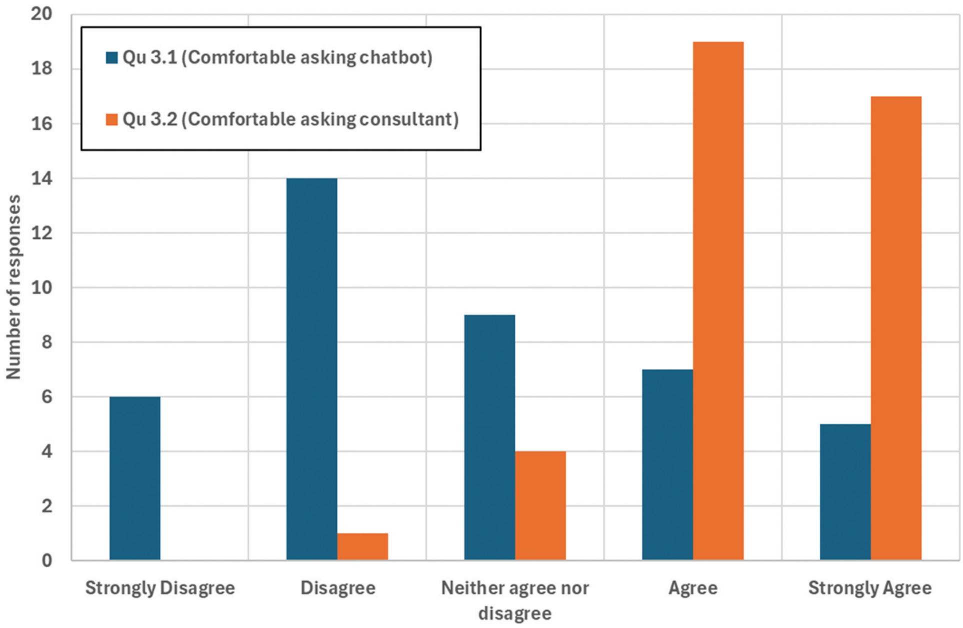

Figure 1. The distribution of responses to question 3.1 "I feel more comfortable asking an AI-consultant chatbot questions (compared to a real consultant)" and question 3.2 "I felt comfortable asking questions during the consultation".

The other theme that was identified in the thematic analysis was 'Prefer Other' ($n = 13$). This was a negative sentiment theme which indicates a preference for engaging in activities other than the AI-consultant or teacher. The codes in this theme were: 'Prefer to consult expert', 'Prefer self-research' and 'Prefer direct practice for defence'. We can see some evidence for this in the response to the Likert question Qu3.1 'I felt more comfortable asking an AI-consultant chatbot questions (compared to a real consultant)'. What is arguably more interesting is the relatively flat response to this question, shown in Figure 1, which highlights that there was a significant cohort of students who expressed a preference for using the AI-chatbots over speaking directly to human consultants or academics. We believe cohorts include students who find interpersonal interactions to be confronting and stressful and prefer anonymous or asynchronous style activities. It is important to note that this is not a strict aversion to in-person meetings as evidenced by the response to Qu3.2 ('I felt comfortable asking questions during the consultation') but a more nuanced preference for the opportunity to sound out ideas with the AI-consultant.

5. Discussion

5.1. Do students perceive a genAI-based teaching activity to be useful?

However, a simple answer to this question could be interpreted from the percentage of the cohort who chose to engage with the learning activity (~86%), which is significant when considering that the activity was not worth any marks and completely voluntary in nature. Despite this, we believe that using the tool does not equate usefulness, instead, to answer this research question we need to consider actual use alongside the sentiment of the cohort combined with their perceived behavioural intent.

Slightly more than half the respondents ($n = 21$) indicated in some way that they found the exercise to be useful, while approximately 25% ($n = 11$) indicated that it was not useful. An analysis of the sentiment of responses shows a close-to-even split, with 80 negative sentiment codes and 69 positive sentiment codes (note that multiple codes may be attributed to each response, and each student responded to two questions). Those that did not find it useful generally cited Performance or Effort Expectancy issues as to the reason why it was not useful. The Performance Expectancy barriers could be easily overcome with a combination of improved design of the learning activities (for example, prescriptive learning activities that use the bot in ways that are reliable) and by better aligning students' perceptions with reality in relation to the information the chatbots produce.

A compounding factor that may be impacting perceptions of usefulness is the level of performance of the student. We did not capture this information in this study, however we can see the possibility that a high achieving student may find the activity to be too low level, or not give a deep enough response to the students' questions. Alternatively, a low achieving student may struggle to ask meaningful questions that prompt the chatbot to go beyond the surface level.

Future work would address this potential moderating factor.

Other factors to consider in these results include the fact that the activities were framed as supporting assessments, and a modest proportion of the cohort (12) had a preference for asking anonymous chatbot questions over humans.

5.2. What specific factors influence voluntary student uptake of a genAI-based teaching activity?

When specifically looking for the barriers to greater voluntary uptake within the framework of the UTAUT, we identified Performance Expectancy as the primary hurdle. Although mirroring the reliability of a human consultant is an important design feature of the AI-consultant in this learning activity (it is not intended to be infallible) concerns about reliability, hallucinations, accuracy and its perceived usefulness, particularly in comparison to other information sources, dominated students' reasoning for not making use of the software.

Minor barriers were also identified in the Effort Expectancy and Facilitating Conditions, but these either overlapped (Effort Expectancy) or could be easily mitigated with better communication and instruction. No social barriers were identified, although students identified that they received more support from peers than from institutions related to the use of genAI.

Enablers were also identified in the thematic analysis again primarily related to the Performance Expectancy. These enablers were more focused on the opportunity to practice and the experience in interacting with the AI, rather than the knowledge or 'correctness' of the tool. Students also highlighted the accessibility and speed of the responses as positive enablers.

Interestingly, the Likert responses were overall much more positive when compared to the written responses. This may be the result of some bias in the students for giving more positive responses or a disconnect between students' evaluation of efficacy/use and their actual experiences in using the tools.

While the tool may be suitable for the intended learning activity, it is ultimately users' *perceptions* of its performance that drive engagement with new technology. A prevalent concern raised by the cohort was the perceived 'unreliability of AI'. Several factors likely contribute to this perception. Firstly, it is widely known that generative AI tools are susceptible to hallucinations – producing inaccurate or fabricated responses – and such instances often dominate discussions around their effectiveness. Secondly, a lack of AI literacy among both staff and students may lead users to inadvertently misuse the tools, resulting in suboptimal prompts and, consequently, misguided conclusions about the AI's reliability. Additionally, the perceived 'unreliability of AI' could be amplified within academic circles or institutions, partly due to the ongoing discourse linking generative AI to risks of academic misconduct. This heightened caution may explain why students rated their peers as more supportive of using generative AI than their institutions. Striking a balance is crucial: while it is important to discourage the misuse of generative AI, it is equally vital to avoid dissuading students from leveraging these tools for legitimate, valuable learning applications.

5.3. Comparison to other studies

Despite the rapid growth of ChatGPT users, and the corresponding growth in papers about ChatGPT, there is currently surprisingly limited work that has emerged on the use of GPT-powered teaching tools within tertiary education. These custom GPT-tools take time to first design, build, and then test with student users, and so there may simply be a time lag in the development and testing of these new tools, that is not associated with ChatGPT itself. In contrast, there is a large body of the literature on the acceptance and use of ChatGPT itself, which we use as a basis for comparison of our own findings.

Within broader population-based studies (Lee et al. 2024; Menon and Shilpa 2023), our findings are broadly consistent, identifying performance expectancy as a major barrier to larger uptake. In a tertiary education context, a UTAUT2 model has been used to study student use of ChatGPT specifically and finds 'performance expectancy', 'habit' and 'Hedonic motivation' as the most dominant positive associations with behavioural intent, which is a finding consistent with our own (Strzelecki 2023). In another related study, a UTAUT model included an additional domain of 'anxiety' and found this to be a barrier to uptake; this result is mirrored in our own work, as we identify a subset of students who expressed a preference for GPT-powered tools (Budhathoki et al. 2024).

We note that all these studies look at ChatGPT specifically, and not GPT-powered custom tools built for specific teaching applications, so there is a limit to the transferability of the findings. Research within the domain of custom GPT-tools for specialised teaching use-cases within tertiary education continues to evolve rapidly, and we expect to see additional studies emerge.

5.4. Limitations & future work

Although the response rate for the survey was acceptable, the small overall cohort size means the study has low statistical power. Additionally, the voluntary nature of the activity and the novelty of the technology would have impacted self-selection into or out of the study. Finally, the analysis methods chosen, while suitable for the task, did not allow for nuance or further investigation as to why a student held specific views.

Future work may expand the study to include formal semi-structured interviews with students or beta testers (including those that opt-out of the activity) to better understand why they engaged or disengaged with the AI tools. Also, it would be good to collect more potential mediating factors from the respondents, such as academic records, prior experience with technology, general aptitude, and other demographic information.

5.4.1. Conclusion

In this study, we used the UTAUT framework to investigate how and why students engage with genAI-based learning activities in the context of a third-year chemical engineering subject. Students were given a survey to complete that included both Likert scale and open-ended questions. A thematic analysis was conducted on the open-ended questions which identified 14 themes related to the four UTAUT constructs.

From this investigation, we found that the Performance Expectancy construct was the most significant barrier and enabler to uptake of the genAI learning tool while Social Influence was not identified in the qualitative responses from students at all. Effort Expectancy and Facilitating Conditions had a minor influence on uptake. The investigation also revealed that a subset of students preferred the AI simulated interactions with a consultant over a human consultant. This highlights the potential for genAI teaching tools to provide alternative forms of access to learning activities for students with diverse needs. These findings will benefit educators looking to implement genAI learning activities into their curriculum.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Dr Chris Honig is a senior lecturer in the Department of Chemical Engineering at the University of Melbourne. His current research focus is on the use of technology to enhance engineering students educational experiences. Dr Honig holds a PhD in Chemical Engineering from the University of Melbourne and BSc from the University of Melbourne.

Dr Shannon Rios is a Senior Lecturer with the Teaching and Learning Laboratory within the Faculty of Engineering and Information Technology. Dr Rios' work within the lab focuses on supporting educators to deliver quality and engaging educational programs by training early career teachers/academics and working with subject coordinators on curriculum review and development projects. Dr Rios has coordinated and taught a wide variety of subjects across engineering and IT including embedded systems, computer engineering, mobile apps and 3D modelling as well as general engineering subjects such as design, maths and physics. Shannon is especially interested in differentiated learning, learning technology, and assessment.

Mr Aditya Desu is a software engineer at the University of Melbourne where he develops AI products to enhance University operations. Mr Desu holds a Masters in Informational Technology (Artificial Intelligence) from the University of Melbourne.

ORCID

Christopher Honig http://orcid.org/0000-0002-4899-6893

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